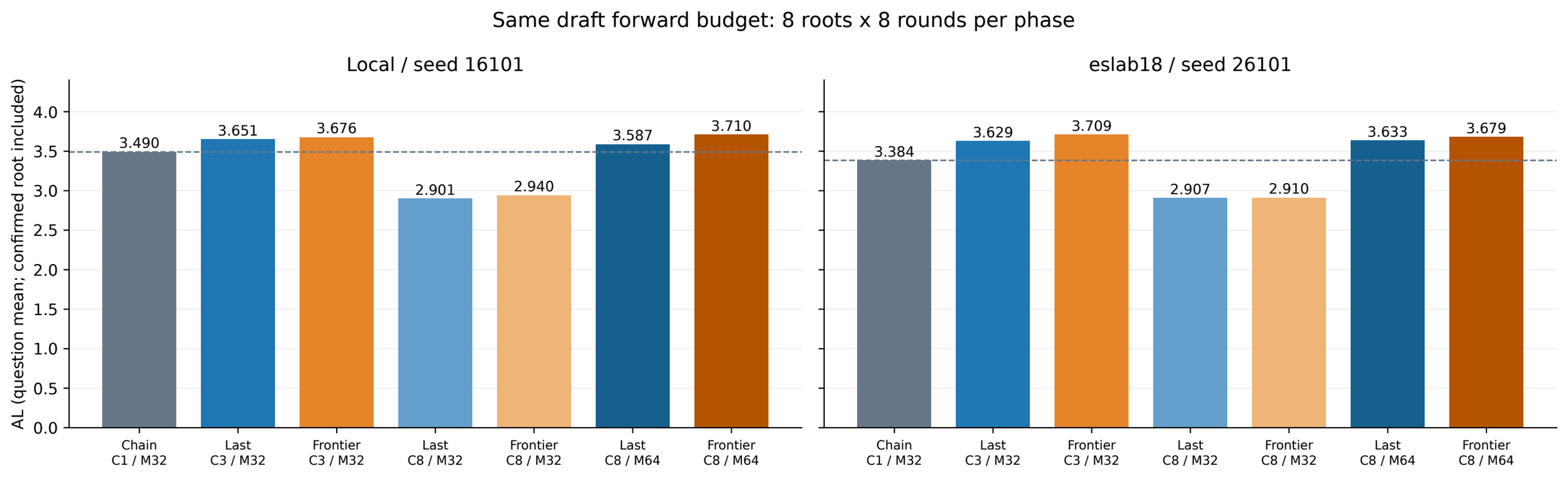
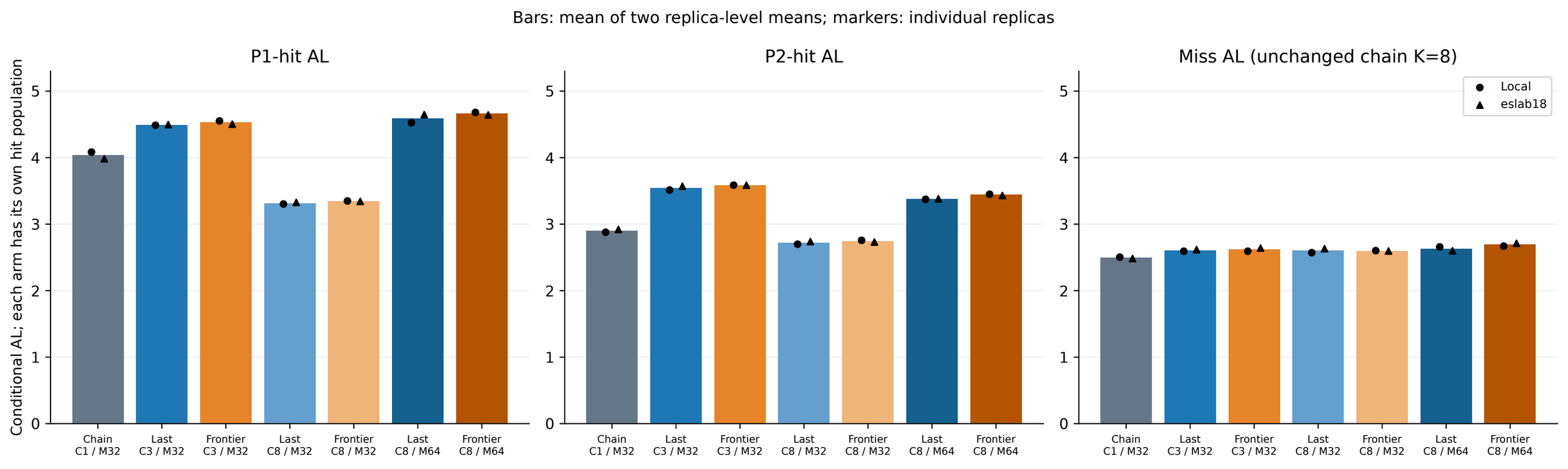
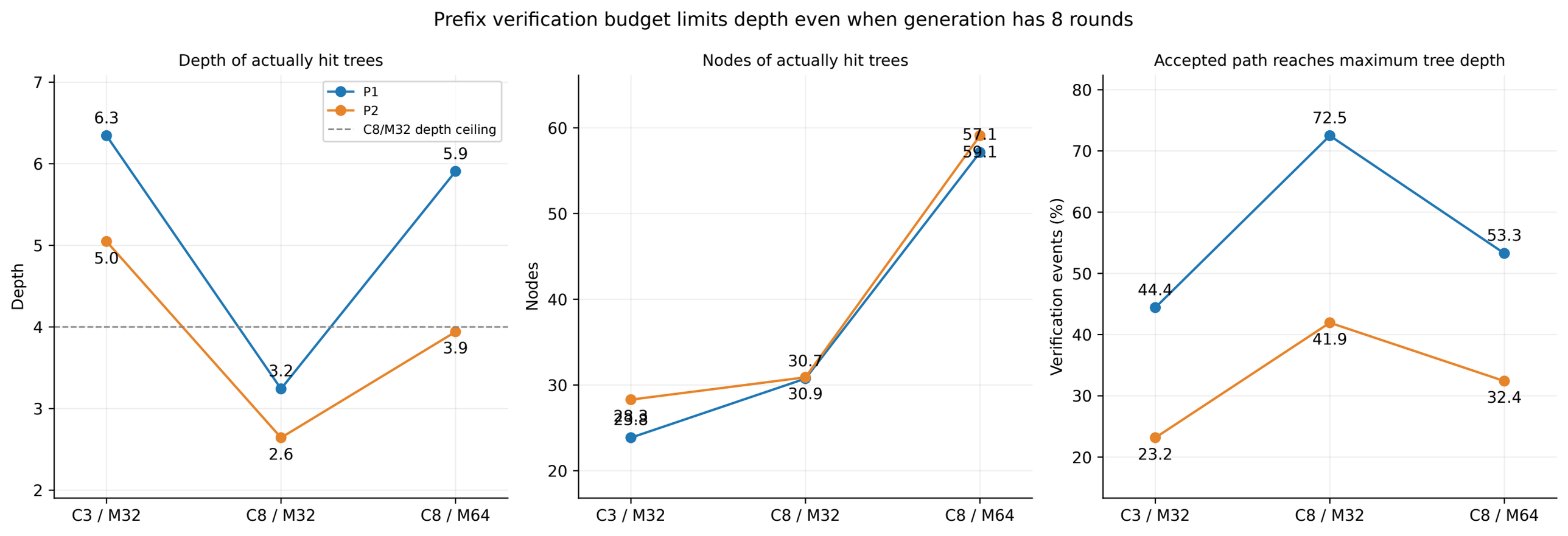
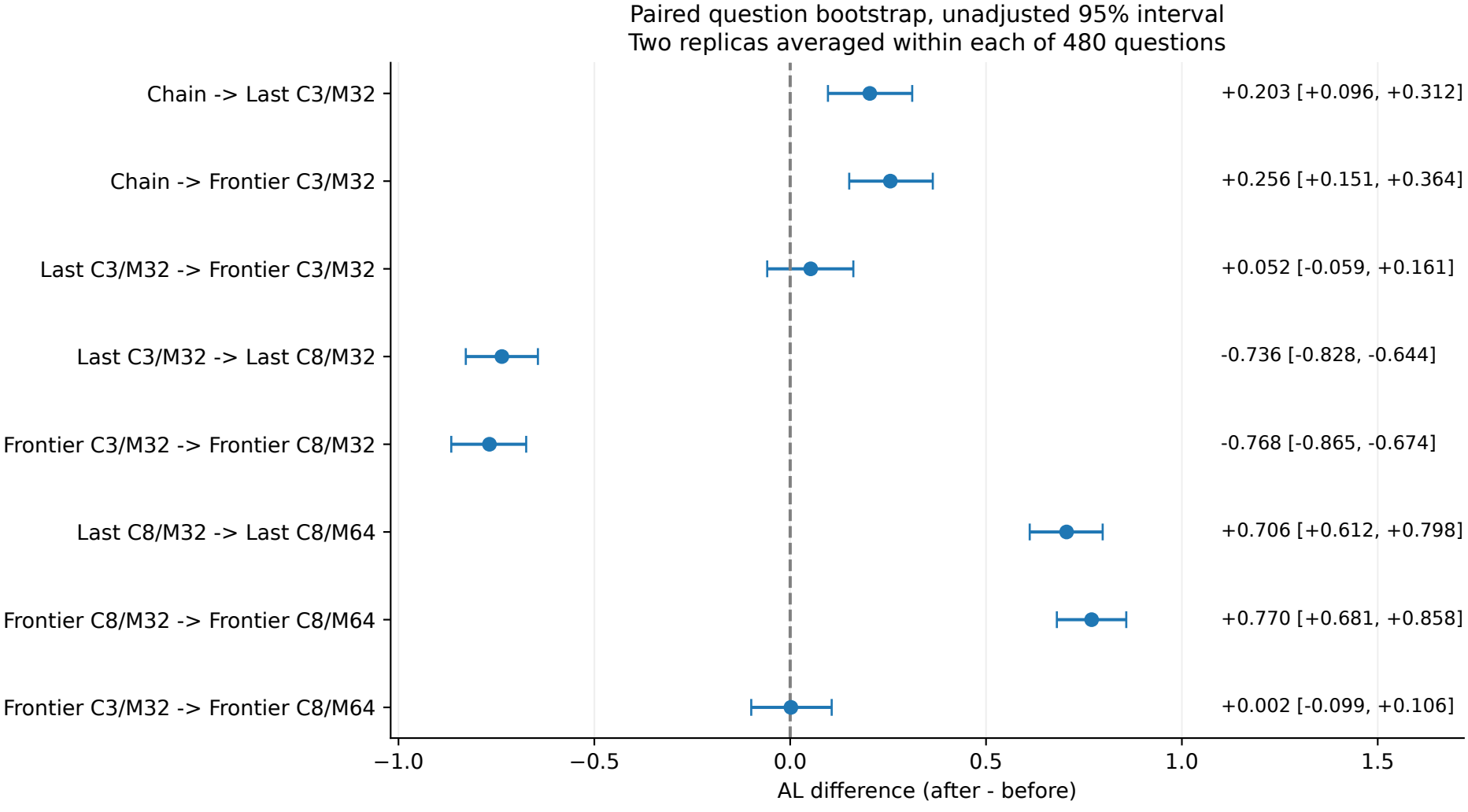
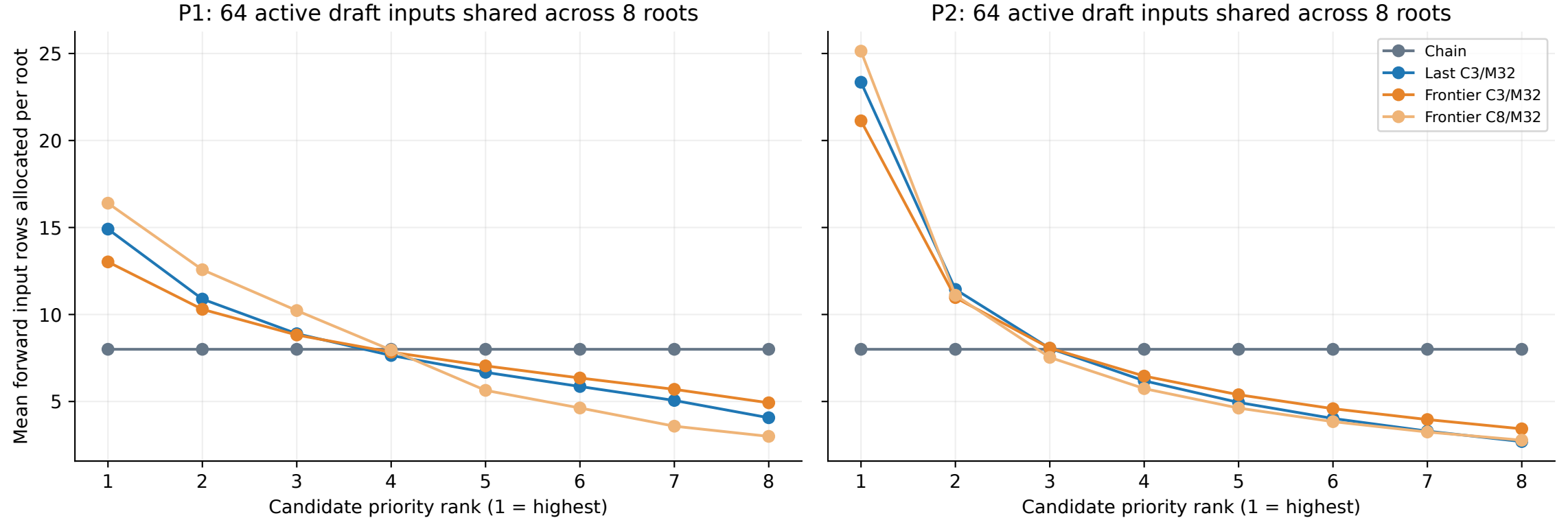

Same draft forward budget: 8 roots x 8 rounds per phase
Local / seed 16101
eslab18 / seed 26101
AL (question mean; confirmed root included)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
3.490
3.651
3.676
2.901
2.940
3.587
3.710
3.384
3.629
3.709
2.907
2.910
3.633
3.679
Chain C1 / M32
Last C3 / M32
Frontier C3 / M32
Last C8 / M32
Frontier C8 / M32
Last C8 / M64
Frontier C8 / M64

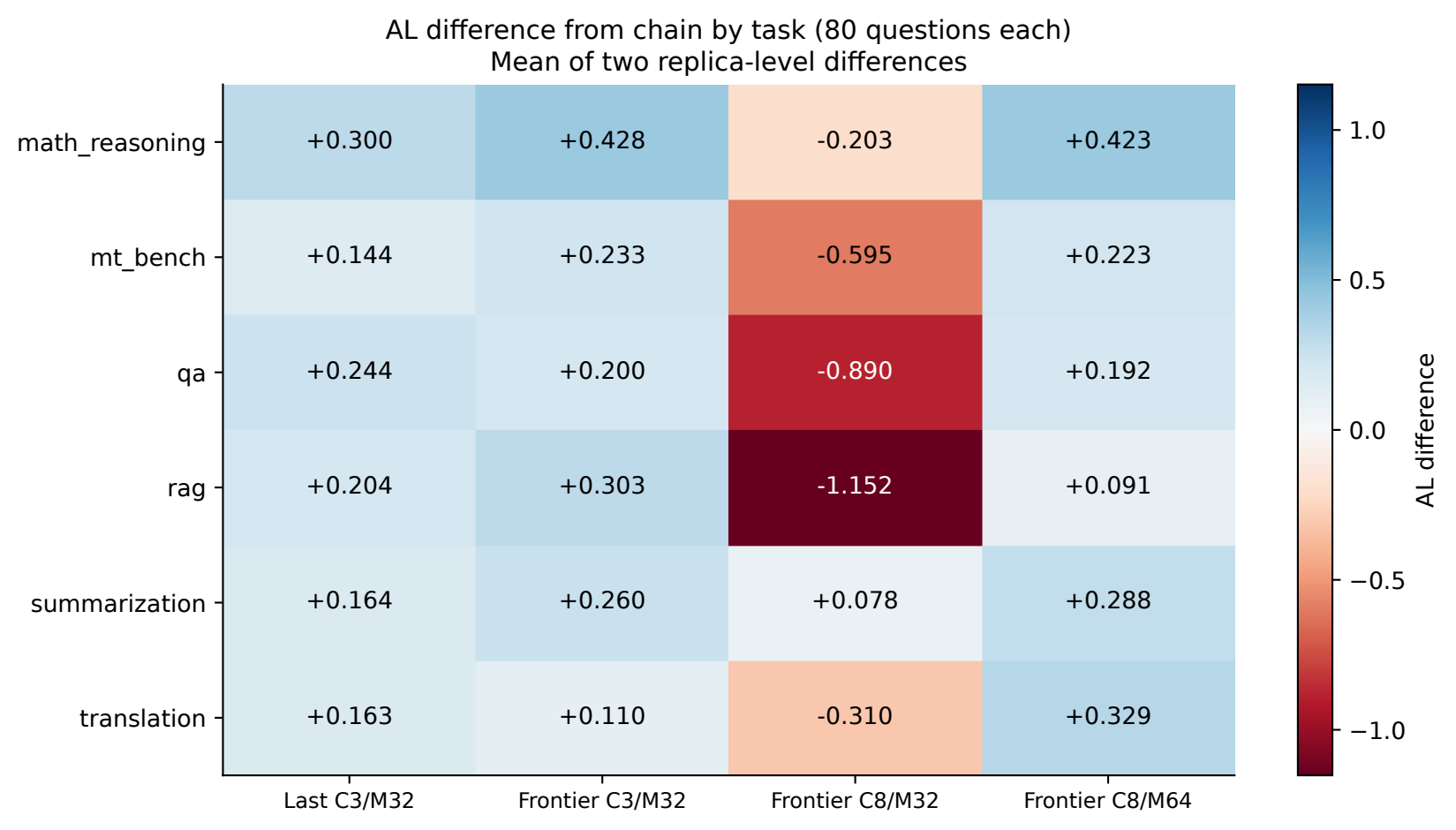

Bars: mean of two replica-level means; markers: individual replicas
P1-hit AL
P2-hit AL
Miss AL (unchanged chain K=8)
Conditional AL; each arm has its own hit population
0
1
2
3
4
5
Local
eslab18
Chain C1 / M32
Last C3 / M32
Frontier C3 / M32
Last C8 / M32
Frontier C8 / M32
Last C8 / M64
Frontier C8 / M64


Prefix verification budget limits depth even when generation has 8 rounds
Depth of actually hit trees
Depth
P1
P2
C8/M32 depth ceiling
6.3
5.0
3.2
2.6
5.9
3.9
C3 / M32
C8 / M32
C8 / M64
Nodes of actually hit trees
Nodes
30.7
30.9
57.1
59.1
Accepted path reaches maximum tree depth
Verification events (%)
44.4
23.2
72.5
41.9
53.3
32.4


Paired question bootstrap, unadjusted 95% interval
Two replicas averaged within each of 480 questions
Chain -> Last C3/M32
Chain -> Frontier C3/M32
Last C3/M32 -> Frontier C3/M32
Last C3/M32 -> Last C8/M32
Frontier C3/M32 -> Frontier C8/M32
Last C8/M32 -> Last C8/M64
Frontier C8/M32 -> Frontier C8/M64
Frontier C3/M32 -> Frontier C8/M64
+0.203 [+0.096, +0.312]
+0.256 [+0.151, +0.364]
+0.052 [-0.059, +0.161]
-0.736 [-0.828, -0.644]
-0.768 [-0.865, -0.674]
+0.706 [+0.612, +0.798]
+0.770 [+0.681, +0.858]
+0.002 [-0.099, +0.106]
−1.0
−0.5
0.0
0.5
1.0
1.5
AL difference (after - before)

All exported forests; not only hit roots. Weighting effectiveness needs a separate ablation.




AL difference from chain by task (80 questions each)
Mean of two replica-level differences
math_reasoning
mt_bench
qa
rag
summarization
translation
+0.300
+0.428
-0.203
+0.423
+0.144
+0.233
-0.595
+0.223
+0.244
+0.200
-0.890
+0.192
+0.204
+0.303
-1.152
+0.091
+0.164
+0.260
+0.078
+0.288
+0.163
+0.110
-0.310
+0.329
Last C3/M32
Frontier C3/M32
Frontier C8/M32
Frontier C8/M64
1.0
0.5
0.0
−0.5
−1.0
AL difference